

## Preparing Bell States from $|00\rangle$ with Quantum Circuits

# Goal: Create Bell states starting from $|00\rangle$

We start with the two-qubit computational basis state

$$|\psi_0\rangle = |00\rangle.$$

Using a small set of gates we can prepare the four Bell states:

$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

Core idea:

- Use a Hadamard on qubit 0 to create a superposition.
- Use an entangling gate (CNOT) to correlate qubit 1 with qubit 0.
- Apply simple single-qubit gates ( $X$ ,  $Z$ ) to obtain the other Bell states.

# Building blocks: single- and two-qubit gates

**Hadamard gate** on one qubit:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

**CNOT** with control qubit 0 and target qubit 1:

$$\text{CNOT} |c, t\rangle = |c, t \oplus c\rangle.$$

Thus, if  $c = 0$  nothing happens; if  $c = 1$ , the target bit flips.

We also use Pauli gates:

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

# Circuit for $|\Phi^+\rangle$ from $|00\rangle$

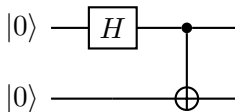
Start with  $|00\rangle$ .

Step 1: Apply  $H$  on qubit 0:

$$(H \otimes I) |00\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle).$$

Step 2: Apply  $\text{CNOT}_{0 \rightarrow 1}$ :

$$\text{CNOT} \frac{|00\rangle + |10\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) = |\Phi^+\rangle.$$



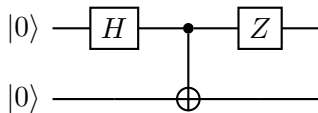
## Circuit for $|\Phi^-\rangle$ from $|00\rangle$

Use the same entangling circuit as for  $|\Phi^+\rangle$ , then add a  $Z$  phase flip.

Since

$$(I \otimes Z) |\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) = |\Phi^-\rangle,$$

we can implement  $|\Phi^-\rangle$  by applying  $Z$  to either qubit (global phase conventions aside).



Equivalent variant (apply  $Z$  on the second qubit):

$$(I \otimes Z) |\Phi^+\rangle = |\Phi^-\rangle.$$

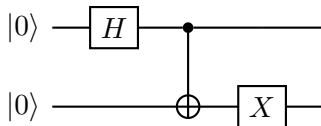
## Circuit for $|\Psi^+\rangle$ from $|00\rangle$

One convenient route:

$$|\Psi^+\rangle = (I \otimes X) |\Phi^+\rangle.$$

Thus: build  $|\Phi^+\rangle$  and then flip the second qubit with  $X$ :

$$(I \otimes X) \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|01\rangle + |10\rangle}{\sqrt{2}} = |\Psi^+\rangle.$$



(You can also apply  $X$  before the CNOT on the target qubit; the above is the clearest construction.)

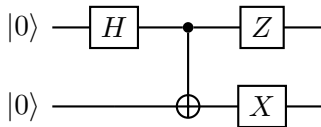
# Circuit for $|\Psi^-\rangle$ from $|00\rangle$

A simple construction uses

$$|\Psi^-\rangle = (I \otimes X) |\Phi^-\rangle \quad \text{or} \quad |\Psi^-\rangle = (I \otimes Z)(I \otimes X) |\Phi^+\rangle.$$

Starting from  $|\Phi^+\rangle$ :

$$|\Phi^+\rangle \xrightarrow{X \text{ on qubit 1}} |\Psi^+\rangle \xrightarrow{Z \text{ on qubit 0 (or 1)}} |\Psi^-\rangle.$$



Up to a global phase, applying  $Z$  on either qubit produces the relative minus sign between  $|01\rangle$  and  $|10\rangle$ .

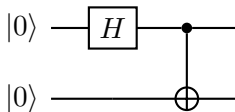
# Summary: One entangling core circuit + local Pauli gates

Core entangling circuit prepares  $|\Phi^+\rangle$ :

$$|00\rangle \xrightarrow{H \otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}_{0 \rightarrow 1}} |\Phi^+\rangle.$$

Then obtain the other Bell states by local operations:

$$|\Phi^-\rangle = (Z \otimes I) |\Phi^+\rangle, \quad |\Psi^+\rangle = (I \otimes X) |\Phi^+\rangle, \quad |\Psi^-\rangle = (Z \otimes I)(I \otimes X) |\Phi^+\rangle.$$





## Optional: Verify with state evolution (conceptual)

If we denote the circuit unitary by  $U$ , then

$$U |00\rangle = |\Phi^+\rangle.$$

Local Pauli operations generate the full Bell basis:

$$\{|\Phi^+\rangle, (Z \otimes I) |\Phi^+\rangle, (I \otimes X) |\Phi^+\rangle, (Z \otimes I)(I \otimes X) |\Phi^+\rangle\}.$$

In quantum information, this is summarized as:

$$\text{Bell basis} = \{(I \otimes \sigma) |\Phi^+\rangle : \sigma \in \{I, X, Z, XZ\}\}.$$